IRWA Final Project - Part 2: Indexing and Evaluation

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GitHub repository: https://github.com/Moca-15/irwa-26.git

Delivery TAG: IRWA-2026-part-2

1. Inverted index and conjunctive search

The system returns only products that contain every query term and ranks them with TF-IDF. The index covers 28.080 products and 20.426 distinct terms.

Each product is one document, identified by its pid. Title, description and product details are pre-processed with the Part 1 pipeline (lower-casing, regex tokenisation, NLTK English stop words, Porter stemming) and merged into one bag of terms. The index maps each term to a posting list {pid: term frequency}, and each product stores the Euclidean norm of its term-frequency vector.

A query is pre-processed the same way. If one query term does not occur in the collection, the result is empty. Matches are scored with idf(t) = log(N / df(t)) and normalised term frequencies.

Test queries

#	Query	Pre-processed terms	AND matches
1	men polo shirt	men, polo, shirt	1.466
2	women printed kurta	women, print, kurta	42
3	blue slim jeans	blue, slim, jean	442
4	women sweatshirt cotton	women, sweatshirt, cotton	513
5	men track pants	men, track, pant	877

2. Evaluation on the validation queries (K = 10)

Metric	Query 1: women full sleeve sweatshirt cotton	Query 2: men slim jeans blue
P@10	1,000	0,600
R@10	0,769	0,600
AP@10	1,000	0,915
F1@10	0,870	0,600
RR@10	1,000	1,000

Metric	Query 1: women full sleeve sweatshirt cotton	Query 2: men slim jeans blue
NDCG@10	1,000	0,706

Metric	Mean over the two queries
MAP@10	0,958
MRR@10	1,000

3. Analysis of the metrics and system limitations

Both validation queries are limited by the top 10 cut-off and by the strict AND filter, more than by the ranking itself.

3.1 What the metrics show

- **Query 1:** the top 10 are all relevant (P@10 = 1,000). R@10 = 0,769 because 13 products are relevant and K = 10 returns at most 10. All 20 labelled products match the AND filter, so the recall loss comes from the cut-off, not from the filter.
- **Query 2:** 6 of the 10 relevant products are retrieved and 4 are not. Three are women's jeans and one is "Tapered Fit Men Blue Jeans", which lacks the word "slim". The AND filter removes all four, so R@10 = 0,600.
- **Two queries** give only two data points. MAP@10 = 0,958 is very sensitive to one query and should not be read as a reliable estimate.

3.2 Evidence from the candidate lists

Top 20 results per test query, from test_query_candidates.csv.

Query	AND matches	Distinct titles in top 20	Most repeated title (copies)	Titles with "Men" / "Women"
men polo shirt	1.466	10	Solid Men Polo Neck Green T-Shirt (4)	20 / 0
women printed kurta	42	10	Women Printed Cotton Blend Straight Kurta (Orange) (6)	0 / 20
blue slim jeans	442	10	Slim Men Blue Jeans (4)	11 / 9
women sweatshirt cotton	513	4	Full Sleeve Printed Women Sweatshirt (10)	0 / 20
men track pants	877	10	Solid Men Grey Track Pants (6)	20 / 0

3.3 Limitations observed

- **Strict AND matching drops relevant products.** A product missing one query term is excluded, even if it matches the rest. A term absent from the catalogue gives an empty result.
- **Near-duplicate products.** "women sweatshirt cotton" has 20 results from only 4 titles. Each copy has its own pid and is judged separately.
- **Gender is not a filter.** "blue slim jeans" has no gender term, so the top 20 mix men's and women's jeans.
- **Inconsistent labels for equal titles.** "Slim Men Blue Jeans" is labelled relevant in two cases and not relevant in one. Relevance depends on attributes the index does not use (colour, fit, price), or the labels are noisy.
- **Single weighting scheme.** Title, description and details share one bag of terms, so a title match counts the same as a match in the details.
- **Small validation set.** Two queries with 20 labelled products each cannot separate systems reliably.

3.4 Proposed fixes

- **Relax the AND filter.** Rank products that match most query terms, and boost those that match all of them. This would recover products like "Tapered Fit Men Blue Jeans". The cost is more noise lower in the list.
- **Remove duplicates.** Group products with the same title, brand, sub-category and price, and show one per group.
- **Weight the fields.** Index title, description and details separately, and give title matches more weight.
- **Turn attributes into filters.** Extract gender, colour and fit from titles (Part 1 fields) and use them as filters or boosts.
- **Use BM25.** It saturates term frequency and normalises document length, which TF-IDF does not.
- **Evaluate on more queries.** Label more queries with graded relevance, and have two annotators label a sample to measure agreement.

4. Relevance judgements for the test queries

The team labelled the top 20 results of each test query as relevant (1) or not relevant (0).

Query	Relevant when (1)	Not relevant when (0)	Relevant in top 20
men polo shirt	Polo or polo-neck T-shirt for men	Plain T-shirt, not polo-neck, or women's garment	20 of 20
women printed kurta	Women's kurta with print, including polka print	Plain kurta, or set with no verifiable print	20 of 20
blue slim jeans	Blue slim jeans, men's or women's	Regular, skinny or tapered jeans that are not slim	13 of 20

Query	Relevant when (1)	Not relevant when (0)	Relevant in top 20
women sweatshirt cotton	Women's sweatshirt made of cotton	Sweatshirt of other material, or no verifiable cotton	20 of 20
men track pants	Men's track pants	Dress pants, jeans or women's garments	20 of 20

How each rule was checked:

Every label comes from a rule applied to the product, not from a visual reading. Title checks match whole words, so "Men" does not match "Women". Where the title cannot show the attribute, the description and product details are checked too. The rules are:

- **Cotton:** the word "cotton" is not in the sweatshirt titles, so it was searched in each product's title, description and details. A product counts as cotton if any of these mention it. All 20 results mention cotton. "Cotton Blend" counts as cotton, because the product contains cotton; a product that is only described as polyester or other material would count as 0.
- **Print for kurta sets:** a product counts as printed if its title says "Printed" or "Print", or if its product details have the field Pattern set to Printed or Polka Print. Kurta sets whose title does not say print are checked in the Pattern field of the product details, which confirms it: all 20 results are printed.
- **Title rules for the other queries:** "men polo shirt" needs "Men" and "Polo Neck" in the title; "women printed kurta" needs "Women" and "Kurta" (print is checked as above); "blue slim jeans" needs "Blue" and "Slim" in the title, for either gender; "men track pants" needs "Men" and "Track Pants" in the title. The sweatshirt rule is the cotton check above, together with "Women" in the title.

Observation: four of five queries are 100 % relevant in the top 20. The AND filter already guarantees the query terms, so the top 20 gives little signal for these queries. This is a limitation of the evaluation, and it is discussed in Section 3.

5. Use of AI tools

We used an AI assistant to help structure this report. We also used it to draft some of the code, but each team member participated, ran it, and can explain each function.